Diabetes Prediction Using Machine Learning

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Swattik Dutta

IDEAS - Institute of Data Engineering, Analytics and Science Foundation

Introduction

During this internship assignment, I worked on a diabetes prediction problem using machine learning techniques. The objective was to analyze a medical dataset and build models that could predict whether a person is likely to have diabetes based on certain health-related measurements.

The dataset used for this project was the Pima Indian Diabetes Dataset, which is widely used for learning and testing classification algorithms. Since diabetes is a major health concern worldwide, I found it interesting to see how machine learning can assist in identifying patterns that may indicate the presence of the disease.

About the Dataset

The dataset contains information about patients such as their glucose level, blood pressure, insulin level, BMI, age, and several other medical attributes.

The target variable in the dataset is **Outcome**:

- 0 – Person does not have diabetes
- 1 – Person has diabetes

Before building any model, I first explored the dataset to understand its structure and the type of information available.

Data Exploration and Preprocessing

The first step was loading the dataset using Pandas and checking its dimensions, column names, and summary statistics.

After understanding the dataset, I separated the independent variables from the target variable. Since the features were measured on different scales, I applied StandardScaler to standardize the data.

This step was particularly important because algorithms such as KNN and SVM are sensitive to differences in feature scales.

The dataset was then divided into training and testing sets using an 80:20 ratio.

Models Implemented

To understand how different machine learning algorithms perform on the same dataset, I trained three different classification models.

Logistic Regression

This was the first model I implemented. Logistic Regression is simple, easy to interpret, and often serves as a baseline model in classification problems.

K-Nearest Neighbors (KNN)

The second model was KNN. This algorithm classifies observations based on the nearest neighboring data points. Although simple in concept, its performance depends heavily on properly scaled data.

Support Vector Machine (SVM)

The third model used was Support Vector Machine. SVM attempts to find the best boundary that separates diabetic and non-diabetic patients.

Model Evaluation

After training the models, I evaluated their performance on the test dataset.

The following metrics were used:

- Accuracy Score
- Classification Report
- F1 Score

- ROC-AUC Score

Using multiple metrics helped provide a more balanced assessment of model performance instead of relying only on accuracy.

Observations

While working on this project, a few observations stood out:

- Feature scaling had a noticeable impact on model performance.
- Logistic Regression produced reliable results and was easy to interpret.
- KNN performed reasonably well after scaling the data.
- SVM showed strong classification capability and handled the dataset effectively.
- ROC-AUC scores helped compare the models more objectively.

One thing I found interesting was that different models approached the same problem in different ways, yet all were able to identify meaningful patterns within the dataset.

Challenges

One challenge was understanding why feature scaling was necessary before applying certain algorithms. Initially, the importance of scaling was not very clear, but after comparing model performance, its impact became evident.

Another challenge was interpreting multiple evaluation metrics and deciding which one should be used for model comparison.

Learning Outcomes

This project helped me gain practical experience in:

- Data preprocessing
- Feature scaling
- Classification algorithms
- Model evaluation techniques
- Working with Python libraries such as Pandas, NumPy, and Scikit-learn

More importantly, it showed how machine learning can be applied to real-world healthcare data.

Conclusion

Through this project, I successfully built and evaluated multiple machine learning models for diabetes prediction. The comparison between Logistic Regression, KNN, and SVM provided valuable insights into how different algorithms behave on the same dataset.

Overall, this assignment strengthened my understanding of the complete machine learning workflow, from data preparation to model evaluation, and gave me hands-on experience in solving a practical classification problem.